

Márk Gyulai

Senior Java Developer

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About me

I build backend systems in Java and Kotlin. I like working on software that stays in use for years and that a team can keep developing with confidence. I also work as an AI consultant, helping companies make artificial intelligence genuinely useful.

Experience

Adroit Group Senior Software Engineer

October 2025 – present

As a senior software engineer I have taken part in three projects, two of them in Kotlin. I develop backends, take part in design and care about quality. One of my projects is about AI consulting.

- **MOL Limo rent a car**: I develop the backend of the MOL Limo car rental system in Kotlin. The backend does not just aggregate data: it also performs calculations and synchronizes asynchronously with a subsystem.
  - Business logic and calculations in the backend, beyond aggregating data
  - Asynchronous synchronization with a subsystem
  - Managing a PostgreSQL database, including transaction management
  - Implementing a custom OAuth flow and token handling with Spring Security
  - Scheduled jobs with Spring's @Scheduled annotation
  - Documentation-first (design-first) approach
  - Writing unit and integration tests, and manual testing
  - Communicating with the client
- **Helvo**: My international team and I built a backend for frontend application for a Swiss fintech startup in Kotlin with Spring Boot 4.
  - Documentation-first (design-first) approach: designing the API in OpenAPI
  - Connecting the backend to other subsystems and the frontend with artifacts generated from the OpenAPI description
- **GLI Solutions – AI Consultant and Developer**: As an AI consultant I help artificial intelligence answer real problems. We find where it is useful, check whether it is feasible, and fit it into the systems that already exist. My team and I also work on the project as developers.
  - Developing a platform that supports manual testing, with a C# and .NET backend, a PostgreSQL database, Node.js and Angular
  - The platform can be connected to AI agents as an MCP server
  - Transaction management in the PostgreSQL database
  - Code-first approach
  - Handling bugs
  - Steering the team and the code towards clean code
  - Implementing repository-level AI skills
  - Writing documentation

Kotlin, Java, .NET, C#, Node.js, Angular, Spring Boot, Spring Security, PostgreSQL, OpenAPI, Exposed, AI, MCP, Prompting

RDI Software Hungary Software Engineer

August 2024 – October 2025

I worked on the McDonald's application, in the team that handles user accounts, authentication and authorization. My team and I developed and maintained microservices and wrote their unit and integration tests. I also wrote SQL queries and connected other systems' APIs.

- **McDonald's**: User accounts, authentication and authorization: I was a member of the international team that handles them.
  - Maintaining existing microservices
  - Designing and developing new microservices
  - Designing API endpoints and tidying up the OpenAPI description file with reusable objects
  - Finding and fixing bugs based on metrics, traces and logs
  - Raising test coverage above 90%
  - Repairing data corrupted by migrations in a NoSQL database with a Python script
  - Managing database transactions
  - Code reviews
  - Implementing Google login with the Google SDK instead of plain API endpoints

Java, Microservices, REST API, OpenAPI, SQL, NoSQL, Python, Authentication, Authorization, Unit tests

EPAM Systems Software Engineer

January 2022 – August 2024

As part of an international team, we developed the financial application of Broadridge, an American company. This is where I really learned multi-threaded programming, domain-driven design and the functional approach. I wrote unit, system and acceptance tests, including UI tests with Selenium. I also connected other systems' APIs, sometimes asynchronously. I solved concurrency bugs, handled database transactions and improved test coverage.

- **Broadridge financial application**
- **Multitenant framework**: The multitenant framework of the Broadridge project, which I worked on as a separate project.

Java, Multithreading, DDD, Multitenancy, Selenium, SQL

EPAM Systems Junior Software Engineer

July 2021 – January 2022

As a junior developer I developed backends and learned a lot along the way. I worked on EPAM's internal exam system and later on the same Broadridge project.

- **Internal exam system**: EPAM's internal exam system: a web application in which exams could be taken safely. My team and I built its backend with Spring Boot 3 and a MySQL database.
  - Designing and developing new features
  - Transaction management in the MySQL database
  - Code reviews
  - Writing unit tests with JUnit
  - Manual testing
  - Improving my skills and learning new technologies
- **Broadridge financial application**: The same project I later worked on as a Software Engineer.

Java, Spring Boot 3, MySQL, JUnit, Unit tests

Skills

AI and MCP | Java 8–25 | Kotlin | Spring Boot | Testing | Tools | Design | Angular

Side projects

lizacover.hu

This portfolio github.com/gyulaimrk/website

Education and certificates

Mathematics and Computer Science, BSc
University of Miskolc, 2017 – 2021

Java Mentored Education Program
EPAM Systems, September 2020 – July 2021

English language exam, B2

Languages
Hungarian, English (B2 language exam)

Driving licence
Category B

Anthropic certificates

Building with the Claude API
Anthropic, March 2026

Introduction to Model Context Protocol
Anthropic, March 2026

Introduction to agent skills
Anthropic, March 2026

Claude Code in Action
Anthropic, March 2026

More certificates (LinkedIn Learning)

Microservices Foundations · June 2021
Docker for Developers · January 2024
Introduction to Docker for Java Developers · January 2024
Java 17: First Look · January 2024
Java: Automated API Testing with REST Assured · January 2024
Learning Java 17 · October 2023
Java 8 for Professionals · October 2023
First Look: Java 10 and Java 11 · October 2020